HW02: Calculus Problem Set

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Instructions. Work neatly and show all steps. You may discuss ideas with a classmate, but write up your own solutions. Scan or photograph your work, combine it into one PDF, name it `lastname_firstname_hw02_problemset.pdf`, and upload it to the HW02 Drive folder (the same one you upload your coding submission to). The problem set is worth **40 points**: 10 points per section.

1 One-Variable Derivatives (10 points)

Problem 1: Differentiation Rules (3 points)

Compute $f'(x)$ for each function. State which rule(s) you use and show all intermediate steps.

(a) $f(x) = (x^2 + 3)^3$

(b) $f(x) = x^3 \ln x$

(c) $f(x) = e^{3x}$

(d) $f(x) = \ln(x^2 + 1)$

(e) $f(x) = \sin(x^2)$

(f) $f(x) = e^x \sin x$

Problem 2: Motion and Rates of Change (3 points)

A ball thrown upward has height (in meters) $h(t) = -5t^2 + 20t$ at time t seconds.

(a) Find the instantaneous velocity $v(t) = h'(t)$.

(b) At what time does the ball reach its highest point? *(Hint: what is $v(t)$ at the peak?)*

(c) What is $h'(0)$, and what does it mean physically?

(d) Find the acceleration $a(t) = h''(t)$ *(the second derivative: the derivative of the derivative — differentiate $h(t)$, then differentiate the result).* What does the sign of $a(t)$ tell you about the motion of the ball?

Problem 3: Nested Functions (4 points)

Compute each derivative. Identify the outer and inner functions before differentiating.

(a) $\dfrac{d}{dx}\, e^{\sin x}$

(b) $\dfrac{d}{dx}\, (\ln x)^3$

(c) $\dfrac{d}{dx}\, \sin(e^x)$

(d) $\dfrac{d}{dx}\, \ln(1 + e^x)$ *(This function appears in deep learning. Can you simplify your answer?)*

2 Partial Derivatives and the Gradient (10 points)

Problem 4: Polynomial Partials (3 points)

For $f(x_1, x_2) = x_1^2\, x_2 + x_2^3$:

(a) Compute $\dfrac{\partial f}{\partial x_1}$ and $\dfrac{\partial f}{\partial x_2}$.

(b) Evaluate both partial derivatives at the point (2, 1).

(c) Write the gradient vector $\nabla f(2,\, 1)$.

Problem 5: Chain Rule with Two Inputs (3 points)

For $g(x_1, x_2) = e^{\,x_1^2 + x_2}$:

(a) Compute $\dfrac{\partial g}{\partial x_1}$ and $\dfrac{\partial g}{\partial x_2}$. *(Treat the other variable as a constant, then apply the chain rule.)*

(b) Evaluate both at (1, 0). Leave your answer in terms of e.

(c) Write the gradient vector $\nabla g(1,\, 0)$.

Problem 6: Choosing a Direction (4 points)

For $f(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_1x_3$:

(a) Compute $\nabla f(x_1, x_2, x_3)$.

(b) Evaluate ∇f at the point (1, 2, 3).

(c) At (1, 2, 3), in which direction should you move to *increase* f as fast as possible? Write the unit vector in that direction. (*Recall: the gradient points uphill.*)

(d) In which direction at (1, 2, 3) does f decrease most quickly?

3 Gradient Descent by Hand (10 points)

Problem 7: Three Coordinates, One Update Rule (10 points)

Let $f(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 4x_3^2$. Starting at the point $(x_1, x_2, x_3) = (3,\ 2,\ 1)$ with learning rate $\alpha = 0.1$:

(a) Compute the gradient $\nabla f(x_1, x_2, x_3) = \left(\dfrac{\partial f}{\partial x_1},\ \dfrac{\partial f}{\partial x_2},\ \dfrac{\partial f}{\partial x_3}\right)$ in terms of x_1, x_2, x_3.

(b) Apply the gradient descent update rule

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \leftarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} - \alpha\, \nabla f(x_1, x_2, x_3)$$

for three steps. Show your arithmetic at each step.

$$\text{Step 0: } (x_1, x_2, x_3) = (3,\ 2,\ 1)$$

(c) What is f at Step 0? Record f after each of your three steps. Is the function decreasing as expected?

(d) Notice that x_1, x_2, and x_3 are all converging toward zero, but at *different rates*. Rank the three components from fastest-converging to slowest. Using the gradient formula from part (a), explain *why* each component converges at a different rate.

(e) Where is the minimum of $f(x_1, x_2, x_3)$? What would happen to each component if you ran gradient descent for many more steps?

4 Chain Rule Through a Computation Graph (10 points)

Problem 8: Updating a Weight and a Bias (10 points)

In machine learning, the loss function is computed step by step, and we need to differentiate through each step. This problem walks you through a mini version of that process — a linear model *with a bias term.*

Suppose we have:

$$z = w \cdot x + b, \qquad L = (z - y)^2$$

where $x = 2$, $y = 5$, and w, b are the parameters we want to learn. We need $\dfrac{\partial L}{\partial w}$ and $\dfrac{\partial L}{\partial b}$.

(a) Compute $\dfrac{\partial L}{\partial z}$ by treating z as the variable. (*Hint:* $L = (z - y)^2$*; use the chain rule.*)

(b) Compute $\dfrac{\partial z}{\partial w}$ and $\dfrac{\partial z}{\partial b}$ separately. (*Treat* $x = 2$ *as a constant.*)

(c) Apply the chain rule to find both partial derivatives of L:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w}, \qquad \frac{\partial L}{\partial b} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial b}$$

(d) Suppose $w = 3$ and $b = 1$. Evaluate z, L, $\dfrac{\partial L}{\partial w}$, and $\dfrac{\partial L}{\partial b}$ numerically.

(e) Take one gradient descent step with $\alpha = 0.05$, updating *both* w and b simultaneously:

$$w \leftarrow w - \alpha \frac{\partial L}{\partial w}, \qquad b \leftarrow b - \alpha \frac{\partial L}{\partial b}$$

What are the new values of w and b? Compute the new loss L. Did it decrease?

Note. The model $z = wx + b$ with $L = (z - y)^2$ is single-weight linear regression with a bias. The chain rule you just applied by hand is exactly what `autograd` does automatically when you call `loss.backward()` in PyTorch.